Initial system

$$\begin{cases} \dot{x}_1 = u \\ \dot{x}_2 = x_1 \left(e^t - 1\right) \\ \dot{x}_3 = x_1^2 \cos\left(t\right) - x_1^2 + x_1 x_2 \end{cases}$$

Series

$$x^0 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \xi_0 + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} \xi_2 + \begin{bmatrix} 0 \\ \frac{1}{6} \\ 0 \end{bmatrix} \xi_3 + \begin{bmatrix} 0 \\ \frac{1}{24} \\ 0 \end{bmatrix} \xi_4 + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \end{bmatrix} \xi_{03} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \end{bmatrix} \xi_{21}$$

Basis Lie elements and their coefficients

$$\begin{aligned}
\mathcal{A}_1 : g_1 &= \xi_0 & v(g_1) &= \begin{bmatrix} -1 & 0 & 0 \end{bmatrix} \\
\mathcal{A}_2 : g_2 &= \xi_1 & v(g_2) &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
\mathcal{A}_3 : g_3 &= \xi_2 & v(g_3) &= \begin{bmatrix} 0 & \frac{1}{2} & 0 \end{bmatrix} \\
g_4 &= [\xi_0, \xi_1] & v(g_4) &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
\mathcal{A}_4 : g_5 &= \xi_3 & v(g_5) &= \begin{bmatrix} 0 & \frac{1}{6} & 0 \end{bmatrix} \\
g_6 &= [\xi_0, \xi_2] & v(g_6) &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
g_7 &= [[\xi_0, \xi_1], \xi_0] & v(g_7) &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
\mathcal{A}_5 : g_8 &= \xi_4 & v(g_8) &= \begin{bmatrix} 0 & \frac{1}{24} & 0 \end{bmatrix} \\
g_9 &= [\xi_0, \xi_3] & v(g_9) &= \begin{bmatrix} 0 & 0 & \frac{1}{2} \end{bmatrix} \\
g_{10} &= [\xi_1, \xi_2] & v(g_{10}) &= \begin{bmatrix} 0 & 0 & -\frac{1}{2} \end{bmatrix} \\
g_{11} &= [[\xi_0, \xi_1], \xi_1] & v(g_{11}) &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
g_{12} &= [[\xi_0, \xi_2], \xi_0] & v(g_{12}) &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
g_{13} &= [[\xi_0, \xi_1], \xi_0], \xi_0] & v(g_{13}) &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}
\end{aligned}$$

Lie elements for principal part

$$\begin{aligned}
\mathcal{A}_1 &: \ell_1 = \xi_0 \\
\mathcal{A}_3 &: \ell_2 = \xi_2 \\
\mathcal{A}_5 &: \ell_3 = [\xi_0, \xi_3]
\end{aligned}$$

The generating set of the Right ideal

$$
\begin{aligned}
&\mathcal{A}_2 : d_1 = \xi_1 && v(d_1) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
&\mathcal{A}_3 : d_2 = [\xi_0, \xi_1] && v(d_2) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
&\mathcal{A}_4 : d_3 = \xi_3 && v(d_3) = \begin{bmatrix} 0 & \frac{1}{6} & 0 \end{bmatrix} \\
&\qquad d_4 = [\xi_0, \xi_2] && v(d_4) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
&\qquad d_5 = [[\xi_0, \xi_1], \xi_0] && v(d_5) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
&\mathcal{A}_5 : d_6 = \xi_4 && v(d_6) = \begin{bmatrix} 0 & \frac{1}{24} & 0 \end{bmatrix} \\
&\qquad d_7 = [[\xi_0, \xi_1], \xi_1] && v(d_7) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
&\qquad d_8 = [[\xi_0, \xi_2], \xi_0] && v(d_8) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
&\qquad d_9 = [[\xi_0, \xi_1], \xi_0], \xi_0] && v(d_9) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\
&\qquad d_{10} = [\xi_0, \xi_3] + [\xi_1, \xi_2] && v(d_{10}) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}
\end{aligned}
$$

Principal part

$$
\begin{aligned}
&\mathcal{A}_1 : \tilde{\ell}_1 = \xi_0 \\
&\mathcal{A}_3 : \tilde{\ell}_2 = \xi_2 \\
&\mathcal{A}_5 : \tilde{\ell}_3 = \frac{\xi_{03}}{2} + \frac{\xi_{21}}{2}
\end{aligned}
$$

Approximating system

$$
\begin{cases}
\dot{x}_1 = -u \\
\dot{x}_2 = -t^2 u \\
\dot{x}_3 = \left(-\dfrac{t^3 x_1}{2} - \dfrac{t x_2}{2} \right) u
\end{cases}
$$

Stationary approximating system

Stationary approximation does not exist